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Organwise Disease Risk Prediction System

A Probabilistic, Organwise Disease Risk Prediction System Using Clinical Categories, Demographic Context, and Lifestyle Factors

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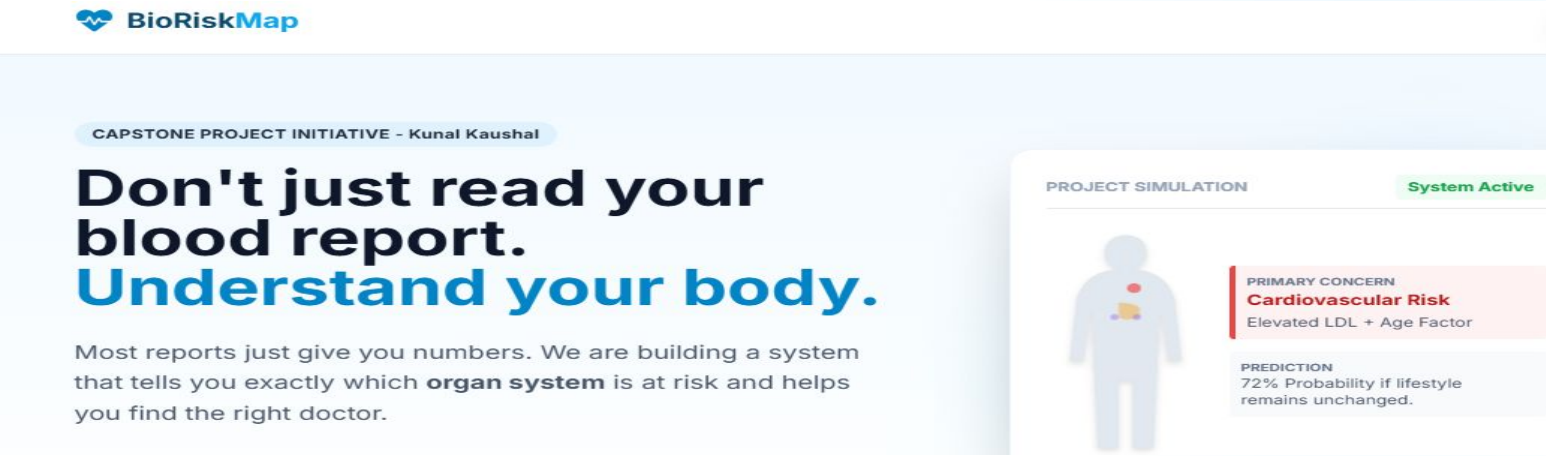
The Problem: Data Without Meaning

- **Volume Overload:** Full-body reports generate massive numerical data but limited actionable insights.
- **Isolation:** Patients see individual biomarkers (e.g., LDL) but don't understand the organ-level impact.
- **Static Reports:** Lack of context regarding age, smoking, and lifestyle modifiers.
- **Delayed Intervention:** High prevalence of lifestyle diseases in India requires early risk stratification.

Which organ is at risk?

Existing systems focus on binary diagnosis. This project aims for **probabilistic screening** to support preventive decision-making before disease onset.

Core Objectives



https://kunalksh.github.io/BioMarkers/#participate

Categorization

Group lab biomarkers into clinically meaningful systems (Cardiovascular, Hepatic, etc.).

Probabilistic Scoring

Compute organ-level risk rather than binary "Yes/No" medical diagnoses.

Context Integration

Adjust risk based on demographic priors like Age, Gender, and Smoking status.

Hierarchical Risk Modeling

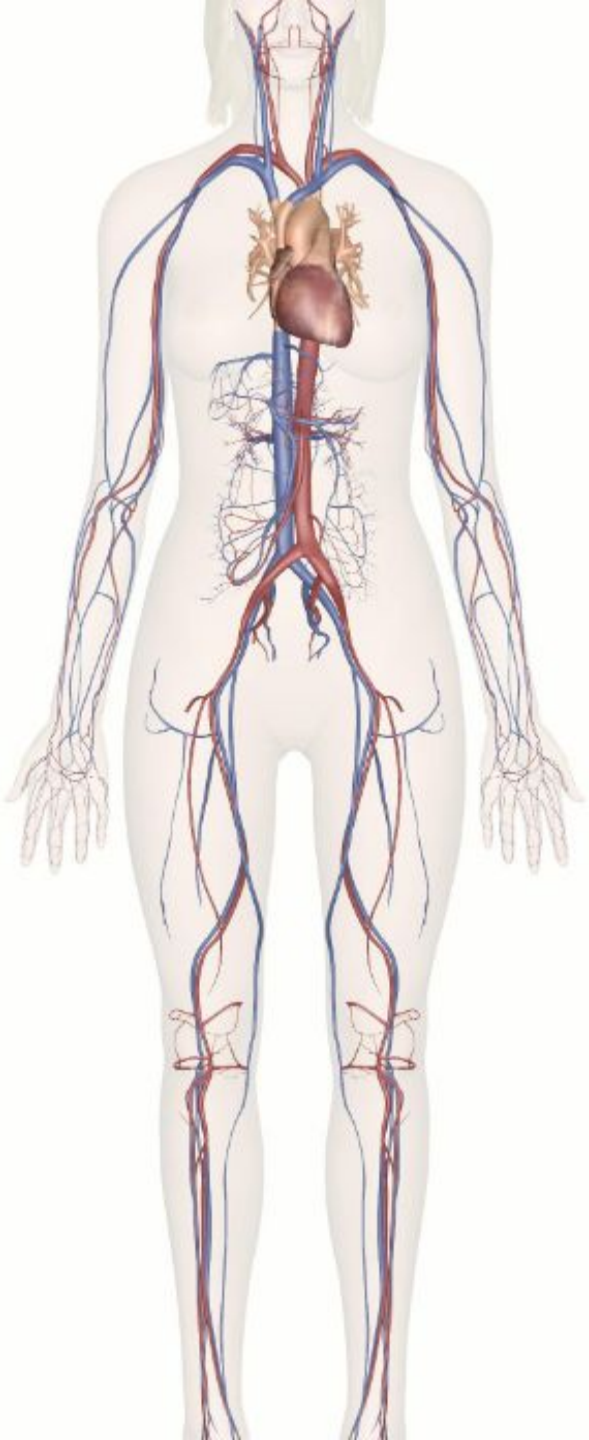
Layer 1: Raw Biomarker Ingestion.

Layer 2: Organ-Specific Aggregation.

Layer 3: Contextual Multipliers (Lifestyle factors).

Layer 4: ML-Based Probability Estimation.

This structure mimics real-world clinical reasoning by looking at "clusters" of evidence.



Current State: Tabular Storage Strategy

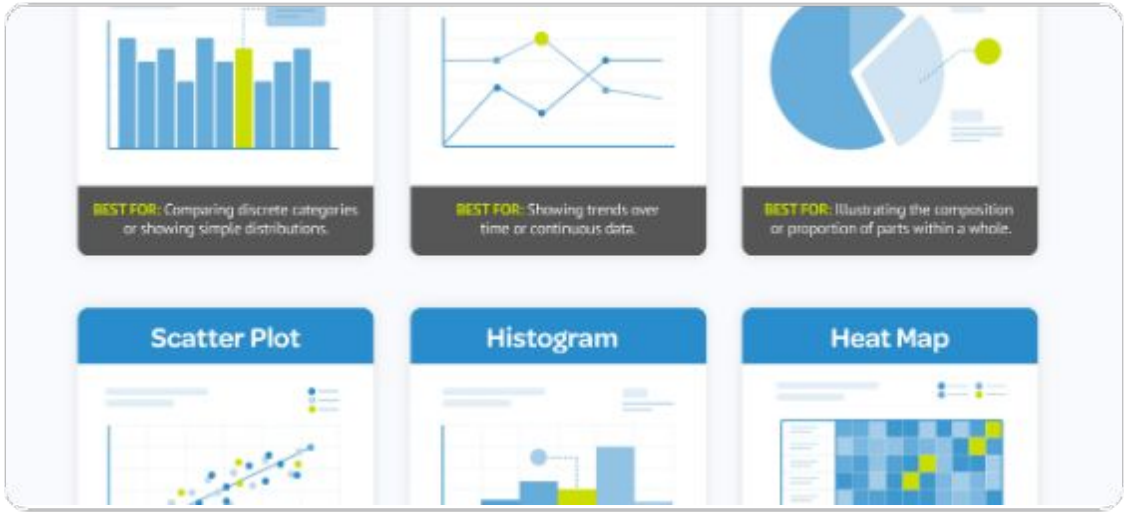
Structured Tabular Persistence

Data is currently maintained in a strictly defined **Relational Schema** using CSV and Excel formats. This ensures immediate accessibility for Python processing engines and high-fidelity data migration.

- **Standardized Headers:** Programmatic cleaning to avoid column name collisions.
- **Type-Strict Columns:** Automatic numeric coercion for computational readiness.

Storage Integration Layer

The system utilizes Pandas and NumPy to import these tables into memory, creating a "System-Wide Global State" that serves as the single source of truth for all organwise risk models.



Future Roadmap: Automated PDF/OCR Ingestion

To eliminate manual data entry, the next phase involves an end-to-end **Ingestion Pipeline** that transforms unstructured medical reports into structured data.

1. Upload Report
2. OCR Extraction
3. Field Mapping
4. Tabular Storage

Technical Stack: PDFReader, Tesseract, Azure Form Recognizer

Integrating PDFReader and Tesseract/Azure Form Recognizer to identify clinical biomarkers and map them directly to our predictive features.

Global Feature Map: 100+ Parameters

The system tracks clinical biomarkers, demographic context, and lifestyle factors mapped to specific physiological domains:

HEMATOLOGY / BONE MARROW

Hemoglobin, RBC, Hematocrit, MCV, MCH, MCHC, RDW, WBC Count, Neutrophils, Lymphocytes, Monocytes, Platelets, ESR.

METABOLIC & PANCREAS

Fasting/PP/Random Sugar, HbA1c, Estimated Avg Glucose (eAG), Serum Insulin, HOMA-IR.

HEART & VESSELS

Cholesterol, LDL, HDL, Triglycerides, VLDL, Non-HDL, Lipoprotein (a), Homocysteine, NT-proBNP.

HEPATIC (LIVER)

SGOT, SGPT, Alkaline Phosphatase, GGT, Bilirubin (T/D/I), Protein, Albumin, Globulin.

RENAL (KIDNEY)

Creatinine, Blood Urea, BUN, Uric Acid, Electrolytes (Na, K, Cl), Calcium, Phosphorus.

ENDOCRINE / REPRODUCTIVE

TSH, T3, T4, Anti-TPO, Cortisol, ACTH, Prolactin, LH, FSH, Testosterone, Estradiol, AMH.

INFLAMMATION / INFECTION

CRP, hs-CRP, Procalcitonin, IL-6, Rheumatoid Factor, Anti-CCP, Anti-dsDNA.

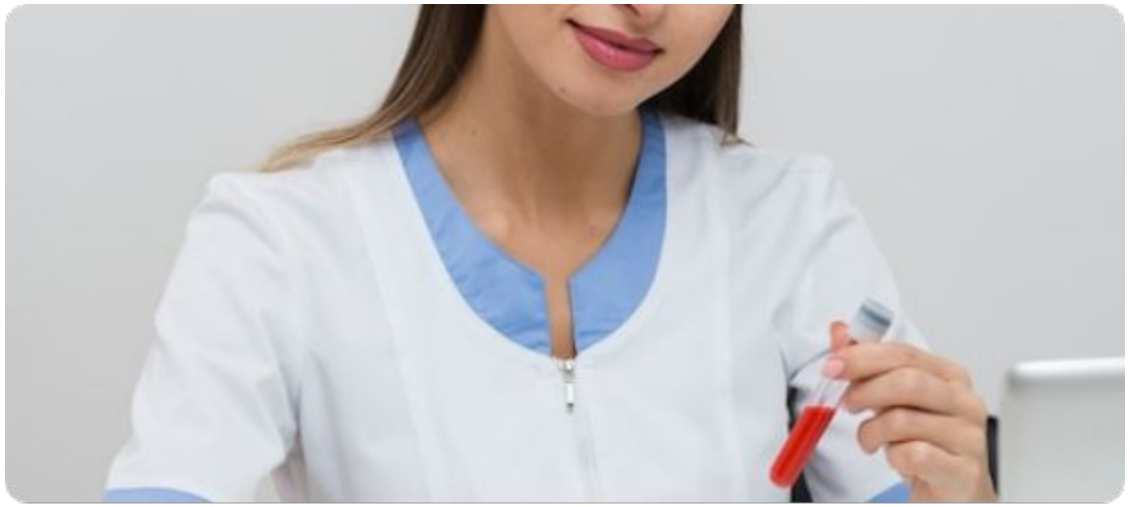
TUMOR MARKERS

PSA, Free PSA, CA-125, CA-19-9, CEA, AFP, Beta-hCG.

LIFESTYLE & METADATA

Patient_ID, Age, Gender, City, Activity, Sleep, Smoking, Alcohol Status.

System Inputs & Variables



Non-Clinical Modifiers

- **Demographics:** Age, Gender, Geography.
- **Temporal:** Date-stamped reports for trend tracking.
- **Lifestyle:** Smoking status, Sleep duration, Physical activity levels.

Clinical Inputs

Biomarkers from Lipid, Liver (LFT), Kidney (KFT), CBC, and Metabolic panels.

Step 1: The Data Ingestion Challenge

Handling Multi-Index Headers

Blood reports often use merged cells for "Organ" and "Biomarker". Our code solves this by programmatically cleaning and flattening headers.

```
clean_cols.append(f"{organ}_{biomarker}")
```

Numerical Transformation

"Yes/No" → 1/0

"Active/Moderate/Sedentary" → 1.0/0.5/0.0

Handling missing values via **SimpleImputer**.

Step 2: Biomarker Categorization

Organ System	Associated Biomarkers (Keywords)
Cardiovascular	Cholesterol, LDL, HDL, Triglyceride, VLDL
Hepatic (Liver)	ALT, AST, Bilirubin, Albumin, Protein
Renal (Kidney)	Creatinine, Urea, BUN, Uric Acid
Hematology	Hemoglobin, RBC, WBC, Platelets
Metabolic	Glucose, HbA1c, Insulin

Code implementation uses a dictionary-based mapping to filter relevant columns dynamically.

Step 3: Calculating Category Stress

Organ Grouping

Biomarkers are dynamically aggregated into clinical clusters (Cardiovascular, Renal, etc.) using automated multi-index header mapping and keyword matching.

Z-Score Abnormality

The system calculates clinical stress by measuring deviations from the population mean for every specific biomarker.

$$z = \frac{x - \mu}{\sigma}$$

Mean Category Stress

Final organ scores are derived from the **Absolute Mean Deviation** of all constituent biomarkers, creating a unified stress index per system.

Step 4: Applying Context

Synthetic Ground Truth

The system classifies "Risk" markers using **Quantile Thresholding** (0.6 quantile) and adds **Gaussian Noise** to labels to prevent overfitting during training.

Age Multiplier

Risks are adjusted by 1% for every year of age. An older patient has a higher baseline physiological risk.

Lifestyle Impact

Cardiovascular risk is amplified by **30%** if the patient is a smoker.

Normalization

The result is a non-linear risk map that considers both the "Lab" and the "Life".

Step 5: Training Strategy

Break-Leakage Methodology

To train models effectively without perfect pattern leakage, we use a custom labeling function:

- ✓ **Noise Injection:** Adds Gaussian noise to breaking mathematical perfection.
- ✓ **Stricter Thresholds:** High-risk labels applied only to the top 40% of scorers (0.6 quantile).

Target Output

Binary labels (0 or 1) are generated per organ to provide ground truth for the classification algorithms.

Soft Voting Classifier

The core engine utilizes a **Voting Ensemble** approach to minimize individual model variance and maximize predictive recall:

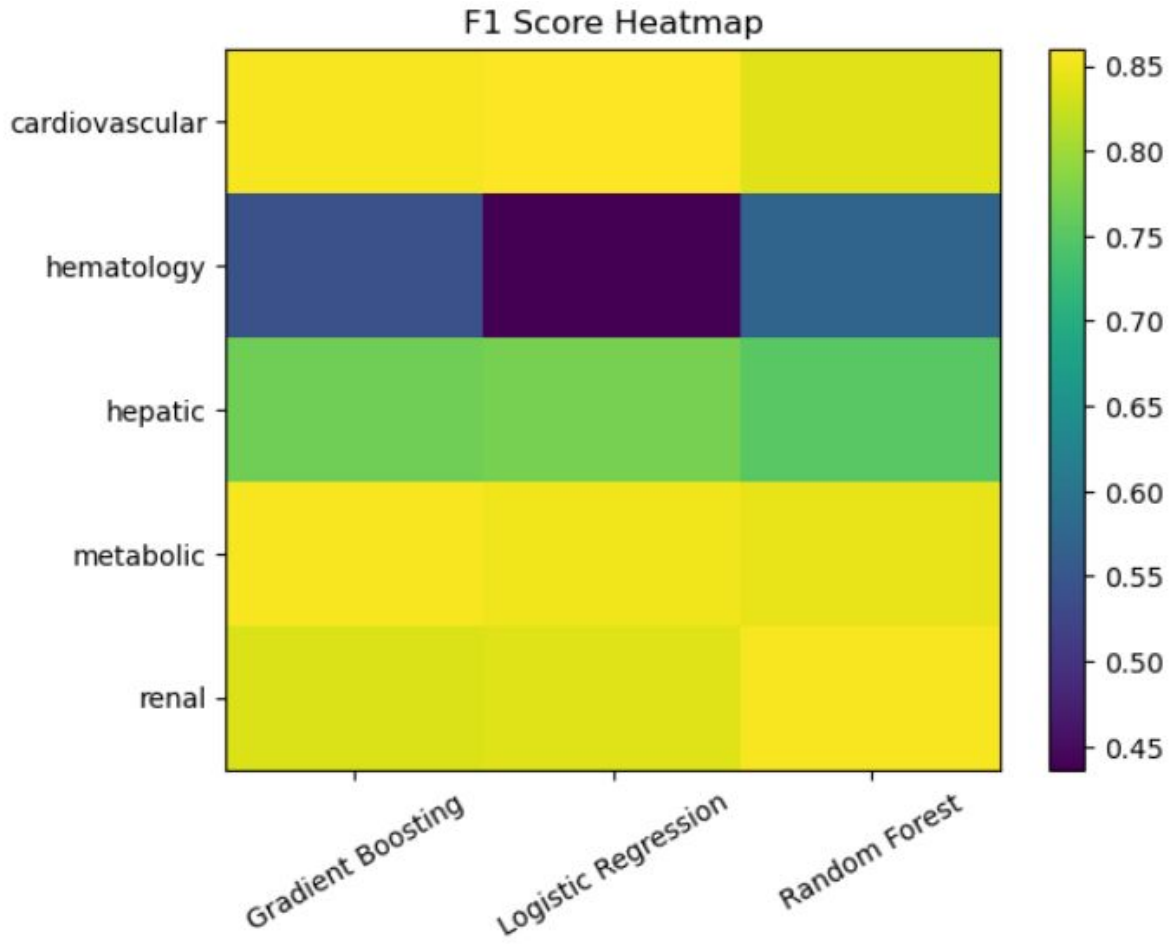
Ensemble Constituents: Logistic Regression + Random Forest + Gradient Boosting.

Risk Classification Tiers

Probability outputs (predict_proba) are mapped to three clinical actionability levels:

HIGH RISK (>70%)	Immediate Clinical Focus
MODERATE (40-70%)	Lifestyle Intervention suggested
LOW RISK (<40%)	Routine Maintenance

Step 6: Comparing Algorithms



Algorithms Tested

Logistic Regression: Efficient for linear relationships in standardized data.

Random Forest: Excellent for capturing non-linear interactions between biomarkers.

Gradient Boosting: High accuracy through sequential correction of errors.

Predictive Engine Analysis

Evaluating Machine Learning Performance for Multimodal Organ-Specific Risk Stratification

Cardiovascular System: Precision Analytics

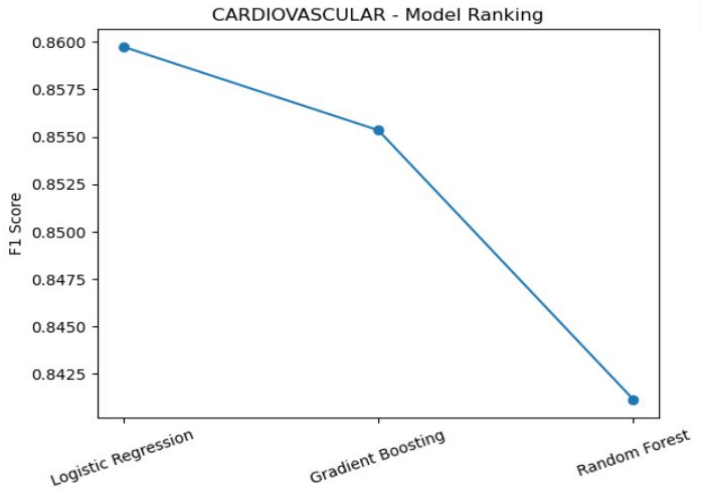
Logistic Regression Performance

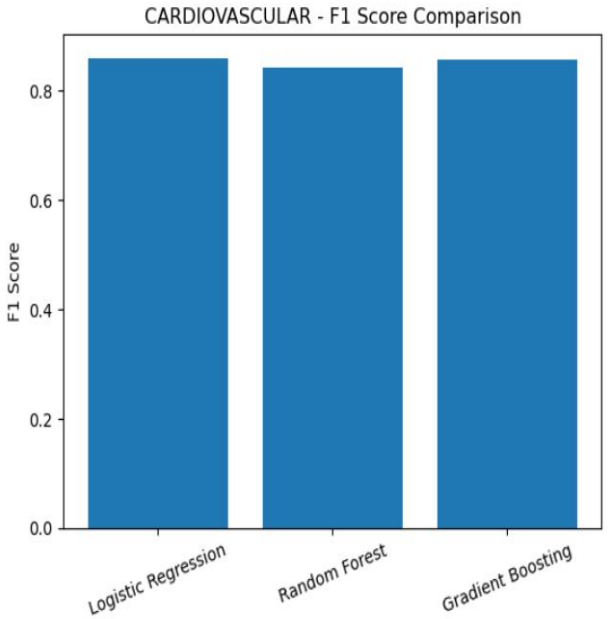
The Cardiovascular engine benefits significantly from linear relationships within lipid biomarkers (Cholesterol, LDL, HDL). Logistic Regression emerged as the most efficient predictor.

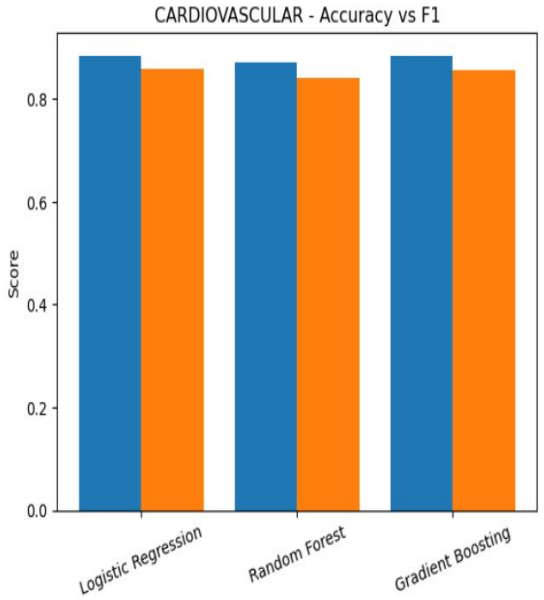
- **Feature Importance:** High sensitivity to LDL/HDL ratios.
- **Reliability:** Consistent performance across age demographics.

0.89
ACCURACY

0.86
F1 SCORE







Hepatic (Liver) Function Evaluation

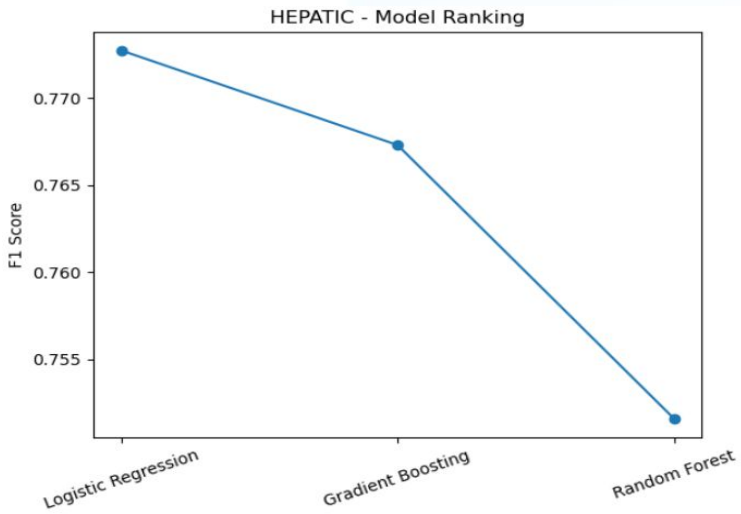
Diagnostic Robustness

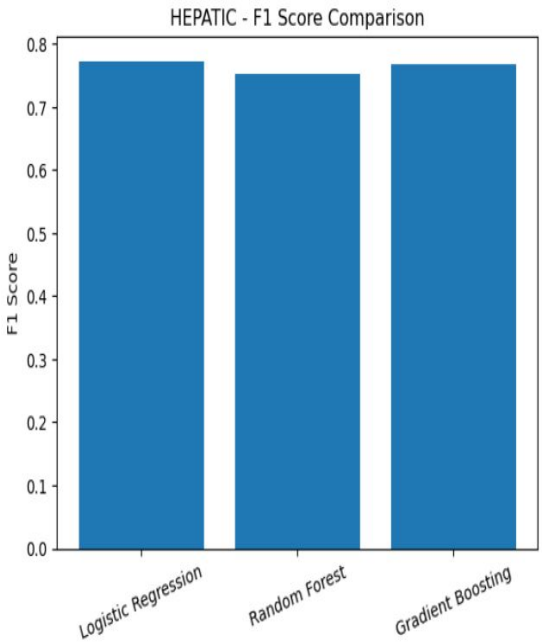
Hepatic risk prediction handles complex enzyme deviations (ALT, AST, Bilirubin). The model effectively manages outlier values in liver function tests.

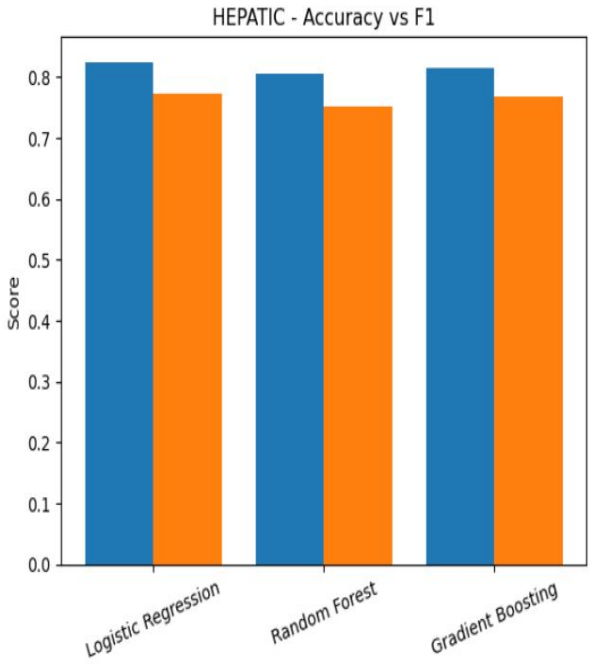
- **Balanced Scoring:** Maintains strong performance despite skewed data distribution.
- **Clinical utility:** High recall for abnormal enzyme spikes.

0.82
ACCURACY

0.77
F1 SCORE







Renal (Kidney) Risk Forecaster

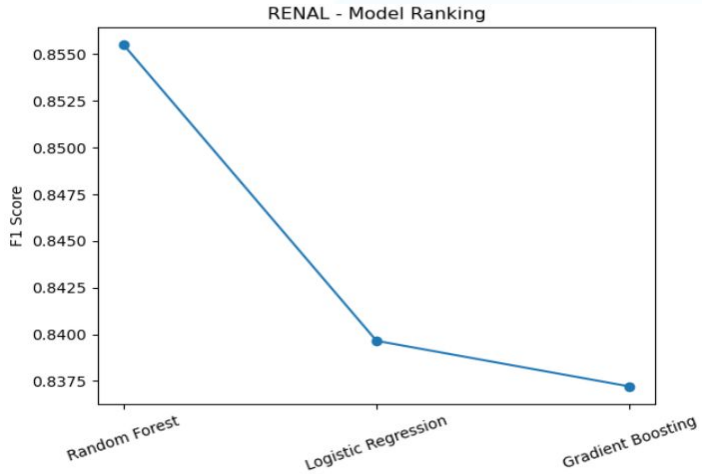
Random Forest Logic

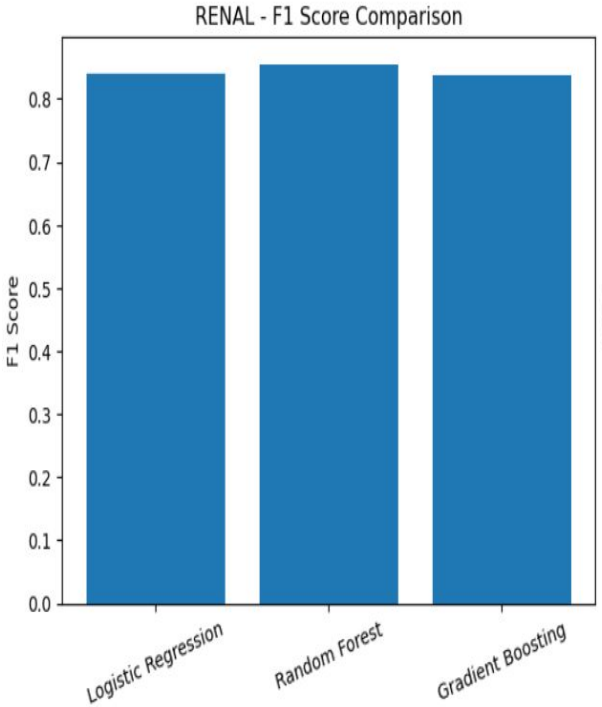
For the Renal system, Random Forest outperformed linear models by successfully capturing nonlinear interactions between Urea and Creatinine levels.

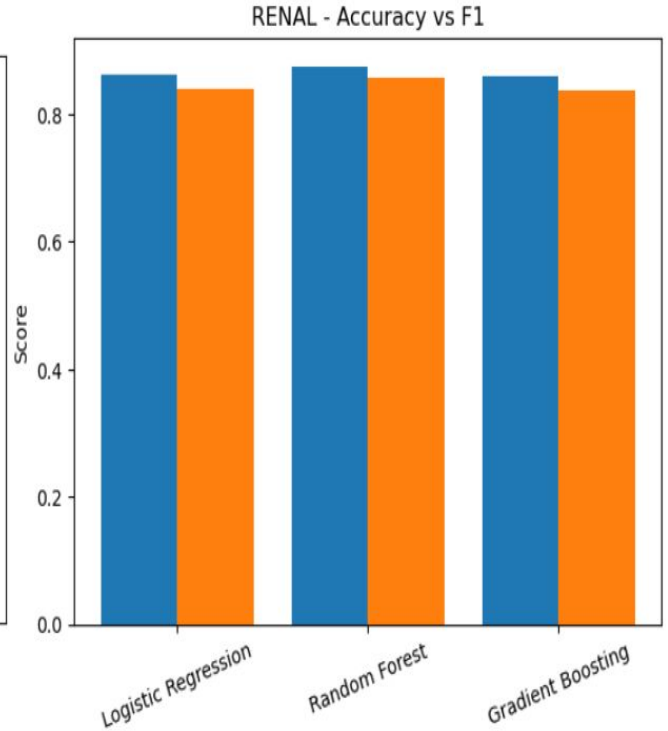
- **Feature Depth:** Effectively learns branching logic for chronic kidney indicators.
- **Risk Calibration:** Precision is prioritized to reduce false positives in high-risk categories.

0.88
ACCURACY

0.86
F1 SCORE







Hematology: Analyzing Complexity

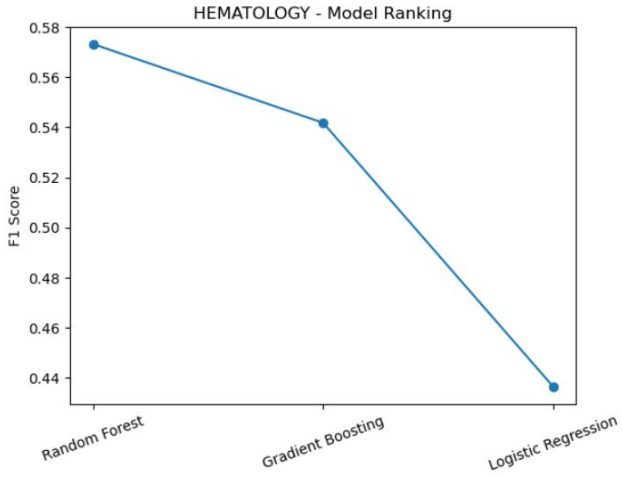
Modeling Challenges

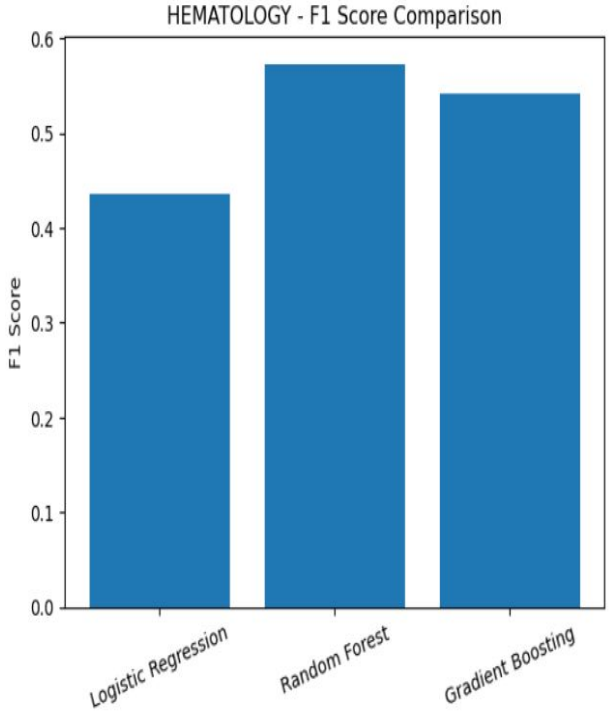
Hematology metrics present the highest modeling complexity. The lower F1 score suggests significant noise in blood cell counts requiring non-linear processing.

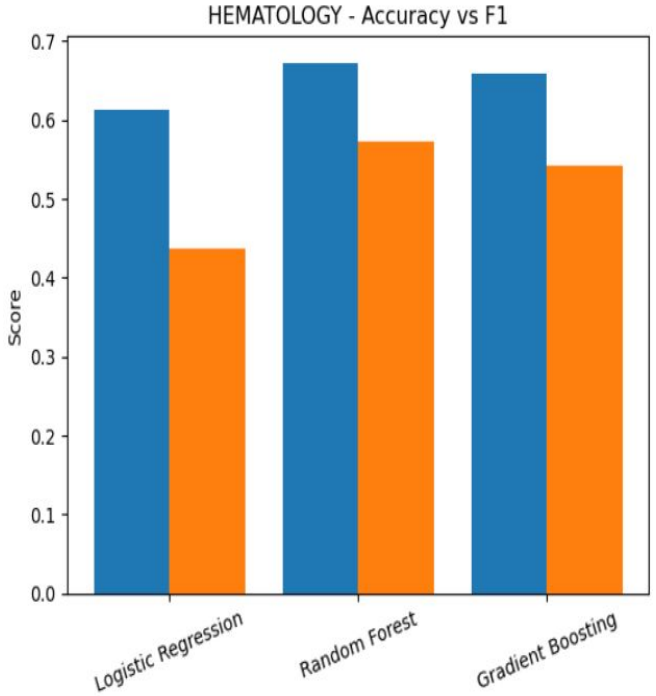
- **Winning Algorithm:** Random Forest provided the most robust baseline.
- **Interpretation:** Low F1 (0.57) indicates a need for more granular longitudinal data.

0.67
ACCURACY

0.57
F1 SCORE







Metabolic Risk Stratification

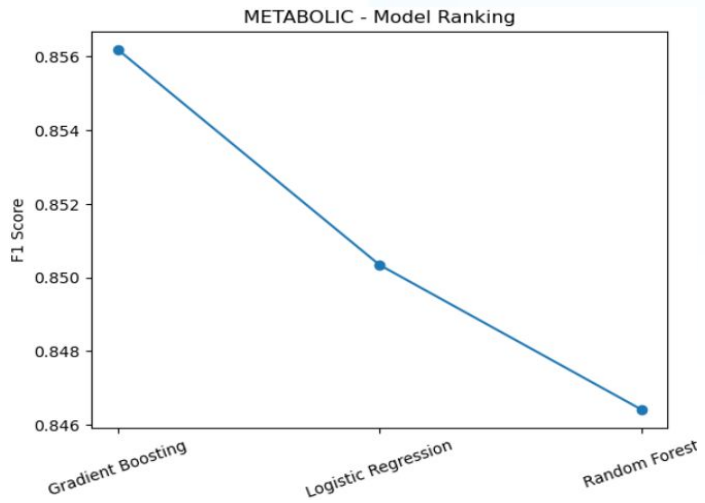
Gradient Boosting Excellence

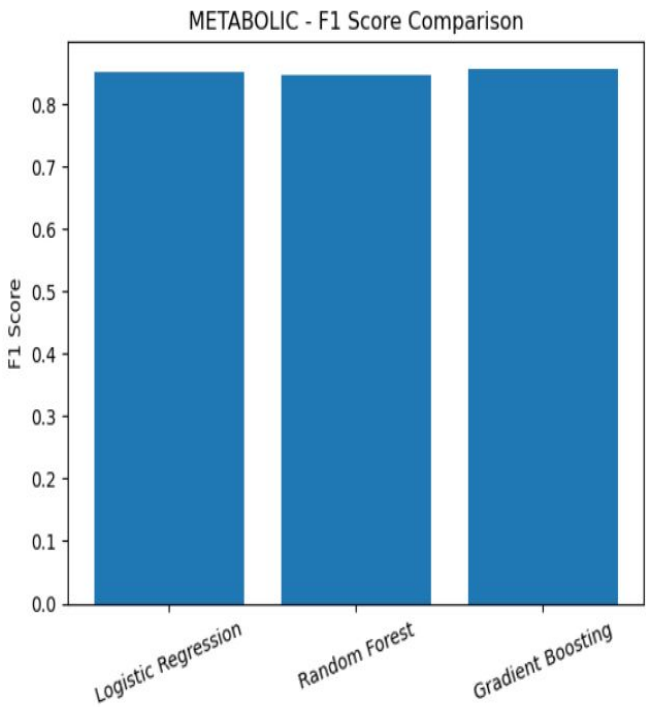
Gradient Boosting emerged as the superior model for metabolic risk, identifying subtle combined trends in Glucose and HbA1c levels.

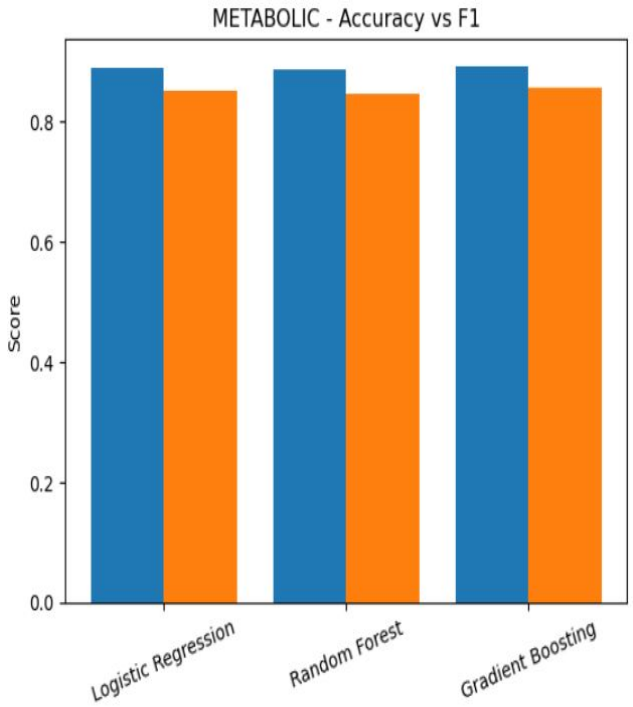
- **Learning Efficiency:** Sequentially corrects errors in risk classification for diabetic markers.
- **Score Stability:** Highest combined accuracy and F1 performance.

0.89
ACCURACY

0.86
F1 SCORE







Global Model Comparison Summary

Final benchmarking of the primary organ-risk predictive models. The scores below represent the "Best Model" selected for final pipeline implementation.

Organ System	Optimal Algorithm	Accuracy	F1 Score	Selection Status
Cardiovascular	Logistic Regression	0.885	0.860	Validated
Hepatic	Logistic Regression	0.825	0.773	Validated
Renal	Random Forest	0.875	0.855	Validated
Hematology	Random Forest	0.673	0.573	Monitor
Metabolic	Gradient Boosting	0.893	0.856	Validated

Insights: Metabolic and Cardiovascular models show the highest predictive confidence, while Hematology requires additional feature engineering.

Recommendation: Use Gradient Boosting as a global backup for systems with non-linear biomarker interactions.

The Challenge: Hematology

Lower Accuracy Metrics

Hematology metrics (CBC) showed the lowest F1-scores (~0.57). This indicates that standard CBC biomarkers have more complex, non-linear relationships compared to metabolic or renal markers.

Lesson Learned: Some organ systems require deeper, specialized biomarkers (e.g., specific cell morphology) for high-confidence risk prediction.

Step 7: The Final Architecture

Ensemble Voting Classifier

Instead of relying on one model, we use a **"Soft Voting"** ensemble of all three top algorithms.

- Reduces variance and over-fitting.
- Captures different signal patterns.
- Provides calibrated probabilities.

Pipeline Logic

SimpleImputer (Mean) → StandardScaler → Soft Voting Classifier

Step 8: Prediction Logic

Probabilistic Output

We extract the **predict_proba[:, 1]** value, converting it to a percentage (0-100%).

Interpretative Tiers

- **< 40%:** Low Risk - Keep monitoring.
- **40-70%:** Moderate Risk - Lifestyle changes.
- **> 70%:** High Risk - Immediate consultation.

The End-to-End Workflow

1. Data Loading

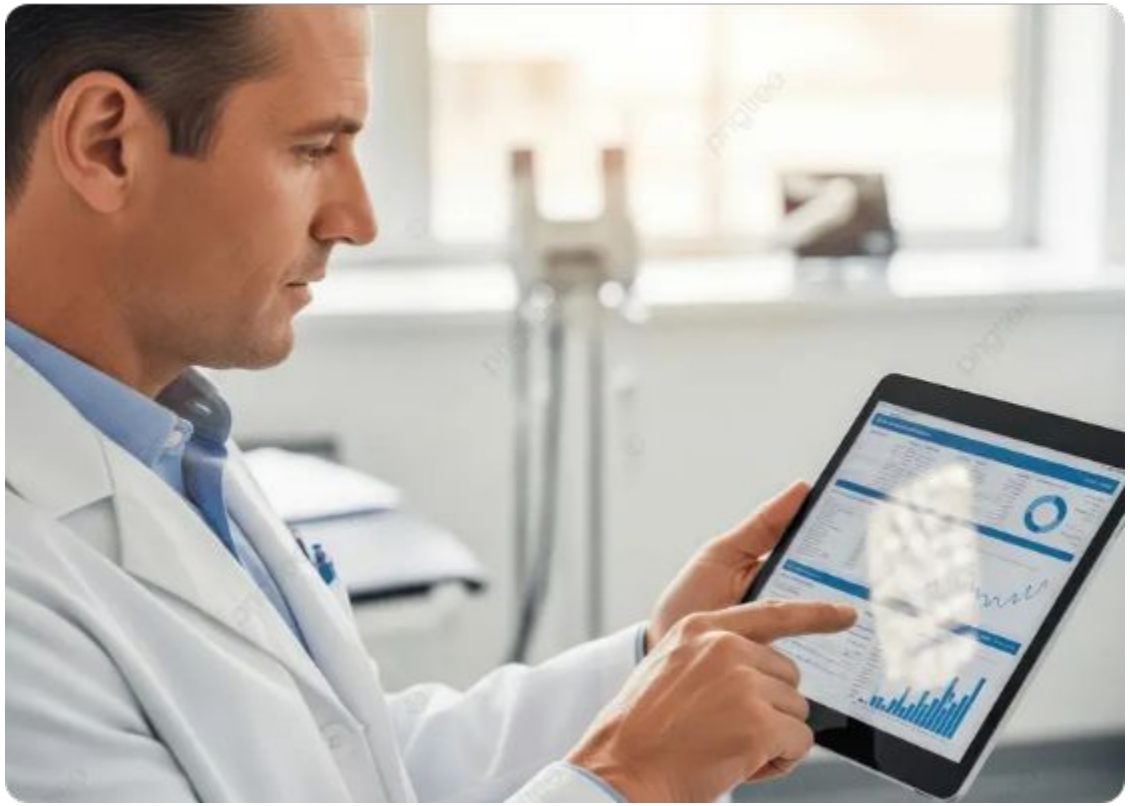
2. Feature Engineering

3. Model Training

4. Patient Risk Inference

The system is designed to break the "Data Leakage" by using only partial feature columns during initial training phases.

Step 10: Patient Risk Profile



The get_patient_risk function

Retrieves individual patient data by ID and generates a detailed report including:

- Demographic Summary (Age, Smoker).
- Organ-by-Organ Risk Percentage.
- Visual Threshold Level (Low/Mod/High).

Case Study: Patient P01

79.72%

RENAL RISK (HIGH)

Patient Profile

Age: 36 **Smoker:** Yes **State:** Telangana

Risk Assessment

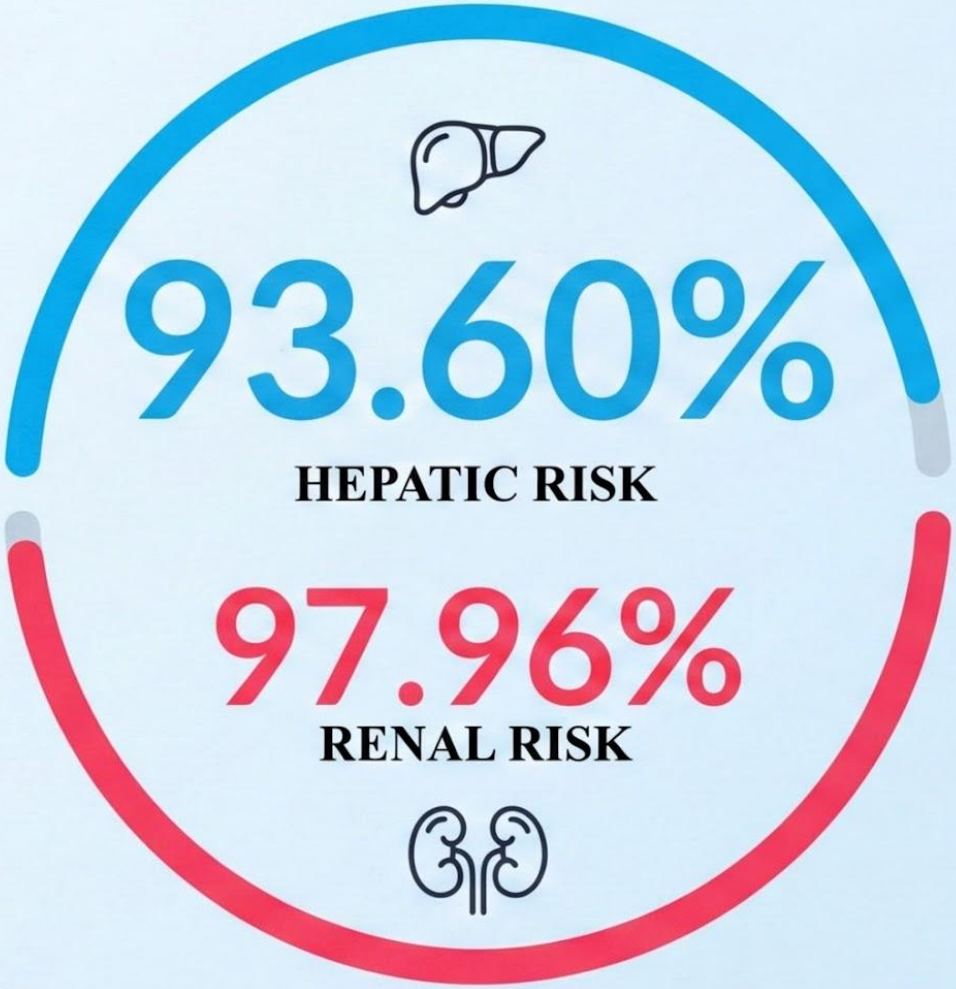
Cardio Risk: 48.65% (Moderate)

Hepatic Risk: 1.01% (Low)

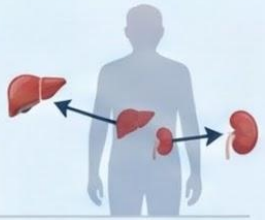
Diagnosis Insight

Despite young age, the smoking context and specific renal biomarkers trigger a high-risk renal warning.

Case Study: Patient P99



Multi-Organ Stress



Age: 69 | **Smoker:** No

This case demonstrates **comorbidity risk.** The system correctly identifies severe hepatic and renal risks simultaneously, reflecting the typical pattern of systemic health decline in senior populations.

Future Tracking: Longitudinal Risk

Risk Trend Analysis

By using **Date-stamped reports**, we can track:

- **Recovery:** Improving scores post-treatment.
- **Progression:** Worsening risk before clinical symptoms.
- **Stability:** Effective management of chronic conditions.

Trend Over Time

A 10% increase in Renal Risk over 6 months is more critical than a single high reading. The system allows tracking via User Identifiers.

Relevance: The Indian Context

Mass Screening

Scalable solution for population health monitoring in high-density urban centers.

Lifestyle Focus

Directly addresses the rising burden of NCDs (Non-Communicable Diseases).

Geographic Priors

Uses city/state data to account for environmental risk factors.

Ethics & Implementation

- **Non-Diagnostic:** The system identifies risk levels, not specific disease names.
- **Anonymization:** No Personally Identifiable Information (PII) is required for prediction.
- **Decision Support:** Intended to empower patients to seek early specialist consultation.

"This system is a tool for awareness and stratification, not a replacement for clinical consultation."

THANK YOU

QUESTION?

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